

This document provides additional assistance with wiring your Extron IP Link® Pro Series enabled product to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific Extron IP Link® Pro Series enabled product or the controlled device manufacturer supplied documentation.

Device Specifications

Device Type:	Display
Manufacturer:	Samsung
Firmware Version:	1066
Model(s):	QN43LS03DAFXZA, QN50LS03DAFXZA, QN55LS03DAFXZA, QN65LS03DAFXZA, QN75LS03DAFXZA, QN85LS03DAFXZA

Tested on the Following Software and Firmware Versions

GC (Plus, Pro) Version	3.24.0.10
IP Link Pro Control Processor Firmware	3.17.0000-b002
MediaLink Plus Controller Firmware	3.04.0000-b001
IP Link Pro Control Processor xi Firmware	1.08.0001-b001
MediaLink Plus Controller Firmware	3.04.0000-b001
HC 400 Series Firmware	2.11.0001-b004
TLP Pro x25 and TLC Pro x26 Series Firmware	3.10.0000-b002
TLP Pro 300M Firmware	1.05.0000-b002

Version History

Driver Version	Date	Notes
1_0_0	6/5/2024	Extron Certified. Initial Version. Tested on model QN43LS03DAFXZA.

Driver Notes

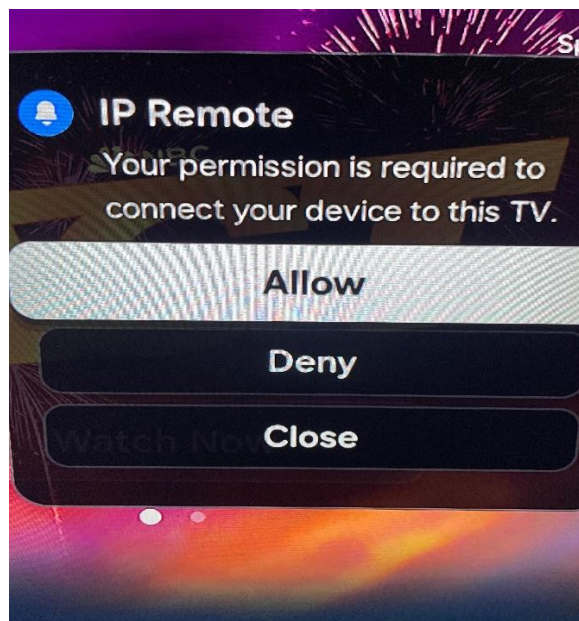
- For serial control
 - User Defined command takes a string parameter and sends the string out of the port.
- For Ethernet control
 - The “SSL Verify Mode” driver parameter (“On” by default) must be set appropriately:
 - “On” if a certificate signed by a trusted Certificate Authority (CA) is loaded to the device.
 - “Off” if a self-signed certificate is loaded to the device.
 - Off will only work with IP Link Control Processor Firmware 3.04.0001-b002 and above.
 - A self-signed certificate may already be loaded by factory default, therefore, Off must be used.

Driver Timings

- Recommended repeat rate for this driver is .1 seconds.

Refresh Token Command

- This device requires a token to be given to the client for Ethernet control to work.
- After the driver is built, use the ‘Refresh Token’ command.
- After this command is used, an ‘IP Remote’ prompt should appear.
- Refer to the image below for reference:



- Use the IR remote to ‘Allow’ the connection.
- The driver will be able to control the display now.

Control Commands & States

Aspect Ratio			
Value	4:3	16:9	Custom ¹
Audio Mute			
Value	Toggle ¹	On ²	Off ²
Channel Step			
Value	Up	Down	
Input			
Value	TV	HDMI 1	HDMI 2
	HDMI 3	HDMI 4	
Keypad			
Value	0 – 9		
Menu Navigation			
Value	Up	Down	Left
	Right	Menu	Enter
	Return	Exit	Home
Multiview Command ²			
Value	None		
Multiview String ²			
Value	String		
Power			
Value	On	Off	
Refresh Token ^{2,3}			
Value	None		
User Defined Command ¹			
Value	None		
User Defined String ¹			
Value	String		
Volume			
Value	0 to 100 in steps of 1		

¹ Only available for serial control.

² Only available for Ethernet control.

³ Please refer to 'Refresh Token Command' section above for details.

Status Available

Aspect Ratio ²			
Value	4:3	16:9	
Audio Mute			
Value	On	Off	
Connection Status ^{RP}			
Value	Connected	Disconnected	
Input			
Value	TV	HDMI 1	HDMI 2
	HDMI 3	HDMI 4	
Multiview String ^{2,3}			
Value	String		
Power			
Value	On	Off	
User Defined String ^{1,3}			
Value	String		
Volume			
Value	0 to 100 in steps of 1		

¹ Only available for serial control.

² Only available for Ethernet control.

³ Emulated status only.

^{RP} Connection Status is based off of the following string:

Serial: \x08\x22\xF0\x00\x00\x00\xE6

Ethernet: URL - https://<IP-Address>:1516

Body - {

"jsonrpc": "2.0",

"method": powerControl,

"params": { "AccessToken": "wByQdCZ3v87ea8UXVrQFNy9qxmywPMfTsO28O7S1QTnRBGCwh"},

"id": "1"}

Cable and Adapter Requirements

Captive Screw to 3.5mm TRS RS-232 Serial Cable

Notes for the Device

Please ensure the following setting is configured for the driver to work. Using the IR remote, open the Menu of the display:

- Settings → Connection → Network → Expert Settings → IP Remote → Enable

Serial communication

Port Type: RS-232

Baud Rate: 9600

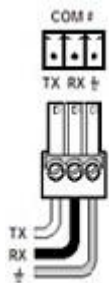
Data Bits: 8

Parity: None

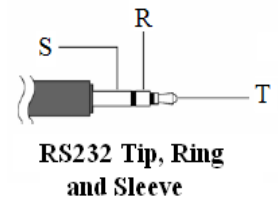
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD		T	TxD
RxD		R	RxD
GND		S	GND



Network communication

When configuring the Ethernet driver, be sure device settings match that of the GCP configuration.

Port Type:	Ethernet (HTTPS)
Default Port:	1516
Logon Credentials	No
Supported:	
Multi-Connection	No
Capabilities:	
Port Changeability:	No

Ethernet Driver Configuration Description

Please refer to user manual for settings and changes to the network communication.

Notes for the Device

Please ensure the following setting is configured for the driver to work. Using the IR remote, open the Menu of the display:

- Settings → Connection → Network → Expert Settings → IP Remote → Enable